Exam 1 Review

Anaya Scott

Question 1

0/1 pt 3 19

In a study, the data you collect is Favorite candy bar.

This data is:

- Quantitative
- Qualitative

Question 2

0/1 pt 3 19

Determine whether the value 55% is a parameter or statistic:

55% of College A's students are women

- Statistic
- Parameter

Question 3

0/1 pt 3 19

Is this value from a discrete or continuous data set.

The number of children a person has

- Discrete
- Continuous

Question 4

0/1 pt 3 19

Does this describe an observational study or an experiment?

A group of students are told to listen to music while taking a test and their results are compared to a group not listening to music

- Experiment
- Observational Study

Question 5

0/1 pt 3 19

Choose the appropriate term for each definition below.

- Sampling bias a measure of how not representative the sample is due to not all members of the population being equally likely to be selected
- Stratified sampling a method for selecting a random sample used to ensure that subgroups of the population are represented adequately; divide the population into groups. Use simple random sampling to identify a proportionate number of individuals from each group.
- Sampling with replace sampling in which every selected member of the population is returned to the population for the future selection
- Interval lev. of measure. a level of measurement using which we can obtain the meaningful difference between the values but there is no natural zero
- Practical signif. a result that is big enough to be meaningful in the real world regardless of its statistical significance

Question 6

0/1 pt 3 19

A doctor wants to know if treatment A or treatment B works faster on removing a wart. He randomly selects a sample of people who had gotten treatment A and a sample of people who have gotten treatment B

Is this an observational study or an experiment?

- ○ experiment
- ● observational study

Question 7

0/1 pt 3 19

Doctors trying to see if a new stent works longer for kidney patients, ask patients if they are willing to have one of two different stents put in. During the procedure the doctor randomly chooses a stent to put in.

Is this a randomized experiment? Why or why not?

- ● yes, since the treatment is randomly selected by the researcher
- ○ no, since the treatment is not randomly selected by the researcher
- ○ yes, since the treatment is chosen by the person in the study

Question 8

0/1 pt 3 19

A business manager wants to see if a new procedure improves the processing time for a task. The manager measures the processing time of a random sample of employees using the old procedure and compares it to the processing time of a random sample of employees using the new system.

Are the two samples matched pairs? Why or why not?

- ● no, two measurements of the same variable have not been taken on the same object or matched objects
- ○ yes, two measurements of the same variable have been taken on different objects that have been matched based on some criteria
- ○ no, two measurements of the same variable have been taken on the same object
- ○ yes, two measurements of the same variable have been taken on the same object
- ○ no, measurements of two different variables were taken on each object

Question 9

0/1 pt 3 19

A doctor has two different weight loss drugs that she usually prescribes. After talking with the patient about the pros and cons of each, the patient decides which one they want to try.

Is this a blind experiment, double blind experiment, or neither? Why?

- ○ blind, since the researcher knows who is getting which treatment but the individual does not
- ○ blind, since the individual knows who is getting which treatment but the researcher does not
- ○ blind, since neither the researcher nor the individual knows who is getting which treatment
- ○ double blind, since both the researcher and the individual know who is getting which treatment
- ● neither, since both the researcher and the individual know who is getting which treatment
- ○ double blind, since neither the researcher nor the individual knows who is getting which treatment

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Question 10 — 0/1 pt ⟲ 3 ⇄ 19

Classify the data as qualitative or quantitative. If qualitative then classify it as ordinal or categorical, and if quantitative then classify it as discrete or continuous.

Variable	Data Type
d Weight (exact)	a. Qualitative Categorical
c Grade (ABCDF)	b. Quantitative Discrete
a Ethnicity	c. Qualitative Ordinal
b Weight (in lbs)	d. Quantitative Continuous

Question 11 — 0/1 pt ⟲ 3 ⇄ 19

Choose the appropriate term for each definition below.

this is generally a union

- (A or B) the event that occurs when A or B or (A and B) occur

- Multi. rule for indep. if events A and B are independent, then P(A and B) = P(A)P(B)

- impossible event event that never occurs

- Probability a measure of how likely something to happen

- Event a collection of outcomes from the sample space

Question 12

0/1 pt 3 19

Choose the appropriate term for each definition below.

- Rand. Variable an unknown quantity whose value depends on chance

- Binomial Exp. a sequence of the same Binomial experiment performed independently n times

- Discrete Random Var. a random variable whose possible values can be listed

- Probability distrib. the table that summarizes the possible values of a discrete random variable and the corresponding probabilities

- Probability histogram a graph that displays the possible values of a discrete random variable on the horizontal axis and the probabilities of those values on the vertical axis

Question 13

0/1 pt 3 19

You want to determine if eating more fruits reduces a person's chance of developing cancer. You watch people over the years and ask them to tell you how many servings of fruit they eat each day. You then record who develops cancer

Is this an observational study or an experiment?

- ○ experiment
- ○ observational cross-sectional study
- ○ observational case-control study
- ● observational cohort study

Question 14

0/1 pt 3 19

A die is rolled twice. What is the probability of showing a 1 on the first roll and an even number on the second roll?

$\frac{1}{6} \times \frac{1}{2} = \frac{1}{12}$

Your answer is : $\frac{1}{12}$

Question 15

0/1 pt 3 19

Two hundred consumers were surveyed about a new brand of snack food, *Crunchicles*. Their age groups and preferences are given in the table.

	18-24	25-34	35-55	55 and over	Total
Liked *Crunchicles*	3	7	3	22	35
Disliked *Crunchicles*	3	4	4	3	14
No Preference	19	24	36	72	151
Total	25	35	43	97	200

One consumer from the survey is selected at random. Use **reduced fractions** for your responses to each of the following questions.

What is the probability that the consumer is 18-24 years of age, given that they dislike *Crunchicles*?

$\frac{3}{14}$ ← 18-24 years ← Total dislike

3/14

What is the probability that the selected consumer dislikes *Crunchicles*?

$\frac{14}{200}$ ← Total dislikes ← Total consumers

14/200

What is the probability that the selected consumer is 35-55 years old or likes *Crunchicles*?

$\frac{(\text{Total } 35-55) + (\text{Total Like}) - \text{Overlap of } (35-55 + \text{Liked})}{\text{Grand Total}}$

$\frac{43+35-3}{200} = \frac{75}{200} = \frac{3}{8}$

3/8

If the selected consumer is 70 years old, what is the probability that they like *Crunchicles*?

$\frac{\text{Liked in 55 over}}{\text{Total 55 over}}$ $\frac{22}{97}$

22/97

● **Question 16** 0/1 pt 3 19

To analyze how Arizona workers ages 16 or older travel to work (carpool, private vehicle, public transportation, other), data from 20 workers were collected and are shown in the table below.

Carpool
Carpool
Carpool
Private
Private
Private
Private
Private
Other
Other
Public
Private
Carpool
Other
Private
Carpool
Public
Other
Carpool
Other

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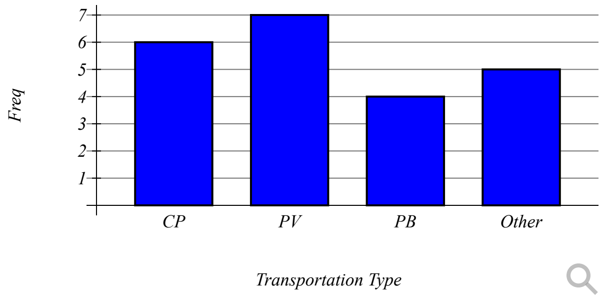
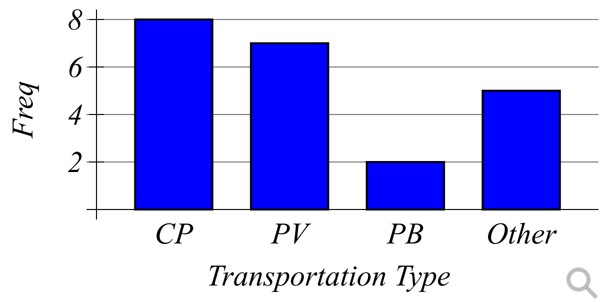
a) Since data were collected for ? 1 Select an answer variable(s), the correct graph to make is a Select an answer bar chart .

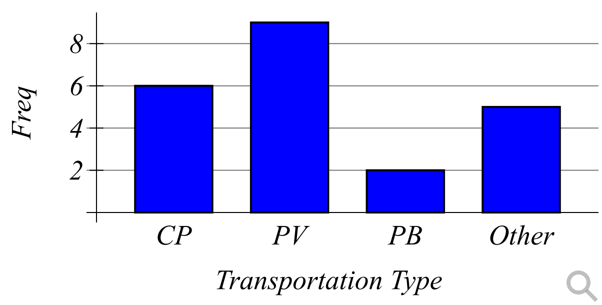
b) Complete the frequency/relative frequency table.

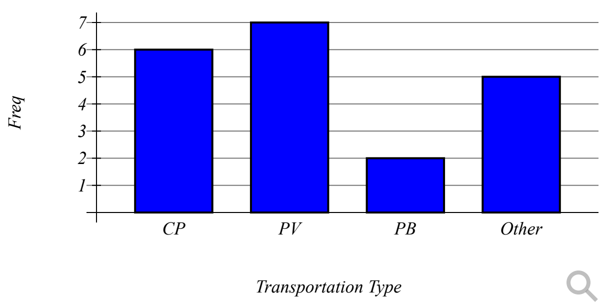
Transportation Type	Frequency	Relative Frequency
Carpool	6	$\frac{6}{20} \to$ 0.30
Private	7	$\frac{7}{20} \to$ 0.35
Public	2	$\frac{2}{20} \to$ 0.10
Other	5	$\frac{5}{20} \to$ 0.25

c) Which of the following is the correct bar chart for the given data?

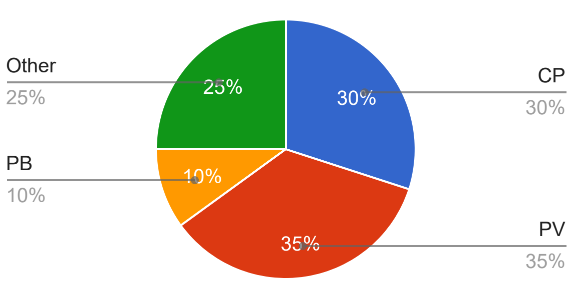


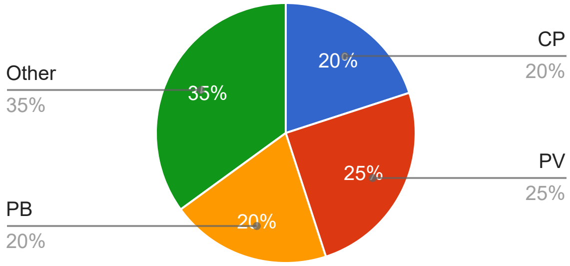


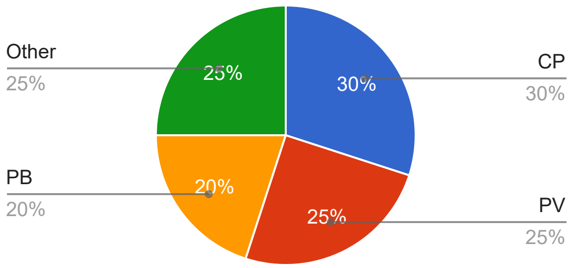


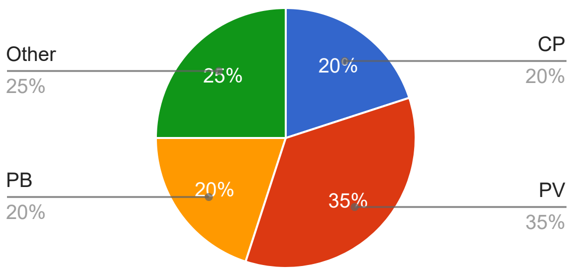


d) Which of the following is the correct pie chart for the given data?







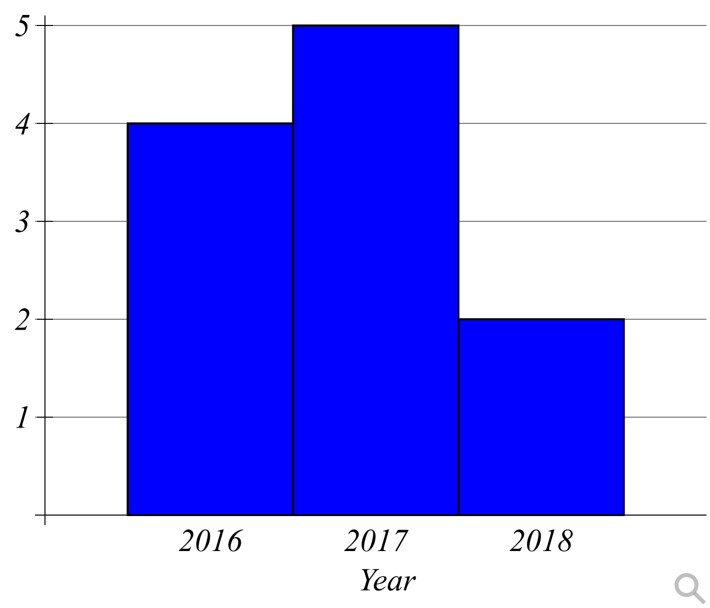


Question 17

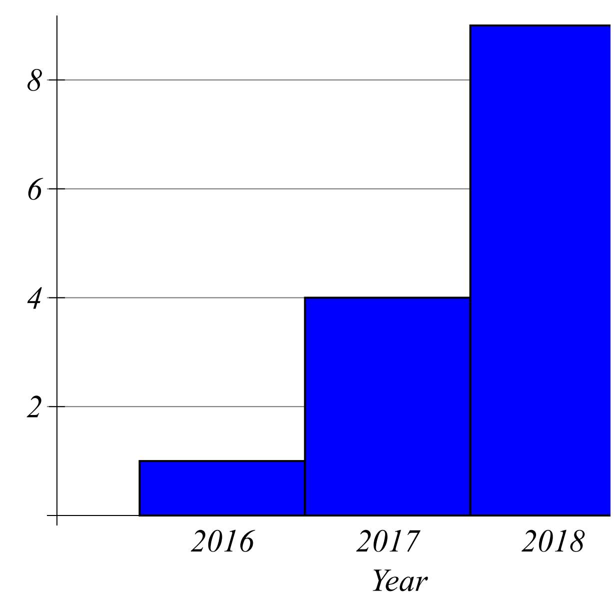
0/1 pt 3 19

The park service records the number of reported rattle snake sightings on Bald Mountain Trail and Camel Back Trail. The data gives rise to the following two bar graphs.

Rattlesnake Sightings on Bald Mountain Trail



Rattlesnake Sightings on Camel Bac



Use the bar graphs to fill in the contingency table below.

Reported Rattle Snake Sightings

	Year			
Trail	2016	2017	2018	Total
Bald Mountain	4	5	2	11
Camel Back	1	4	9	14
Total	5	9	11	25

Question 18 0/1 pt 3 19

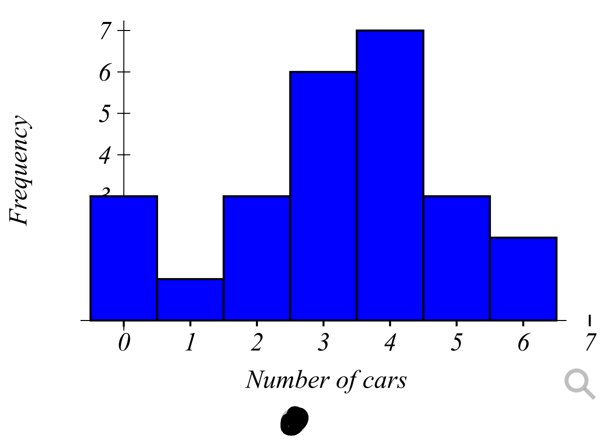
Tomás has surveyed their 25 classmates about the number of cars in their household and received the following data:

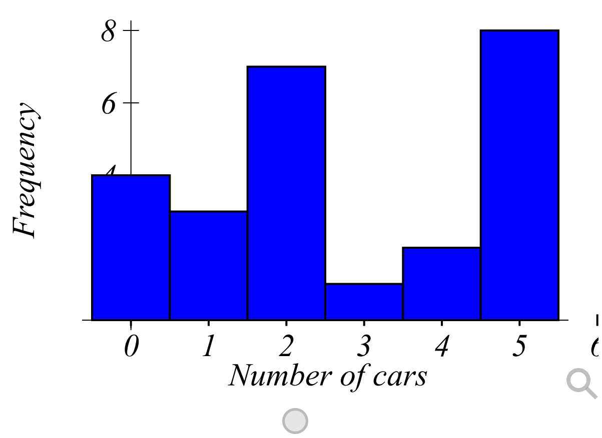
0	2	1	5	3
4	3	3	5	4
4	2	3	4	4
2	3	0	0	3
4	5	4	6	6

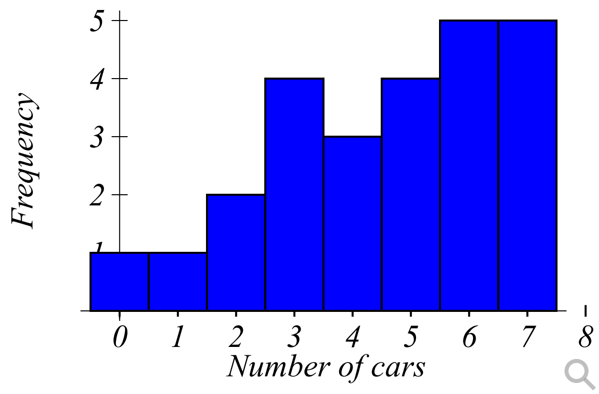
Construct a frequency and relative frequency distribution table: (Round relative frequencies to 2 decimal places)

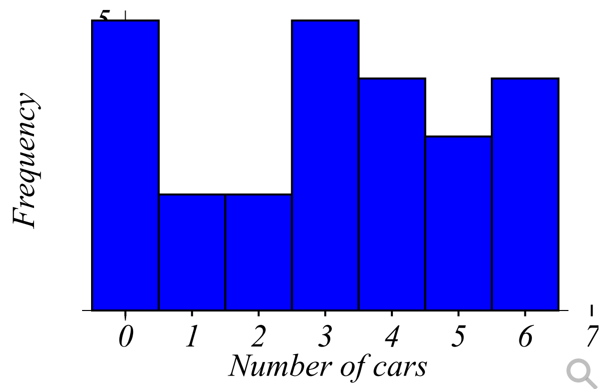
Number of cars	Frequency	Relative Frequency
0	3	0.12
1	1	0.04
2	3	0.12
3	6	0.24
4	7	0.28
5	3	0.12
6	2	0.08
Total:	25	1.00

Construct a histogram:









Question 19

0/1 pt 3 19

Kelsey surveyed 16 students and recorded the following data:

13	34	38	17	30
33	20	22	35	38
18	23	26	9	32
25				

Construct stem-and-lead diagram. Don't use commas (so like 1,2,3 is 123).

0| 9

1| 378

2| 0256

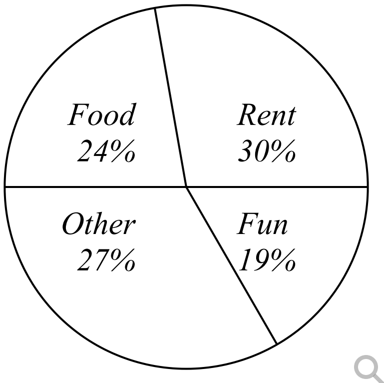
3| 02334588

4|

Question 20

0/1 pt 3 19

Carie categorized her spending for this month into four categories: Rent, Food, Fun, and Other. The percents she spent in each category are pictured here.



If Carie spent a total of $2000 this month, how much did she spend on Fun?

- ○ $360
- ○ $19
- ● $380

Question 21

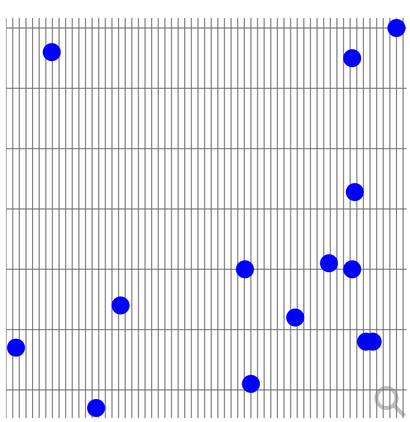
0/1 pt 3 19

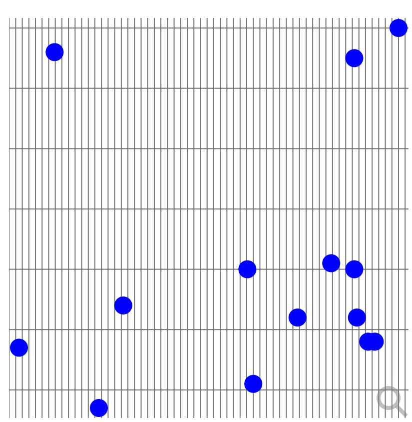
The World Bank collects information on the fertility rate (children per woman) in the country and the life expectancy of a person (in years) in each country ('Life expectancy at,' 2013 and 'Fertility rate,' 2013). The data for 14 randomly selected countries for the year 2011 are in the table below.

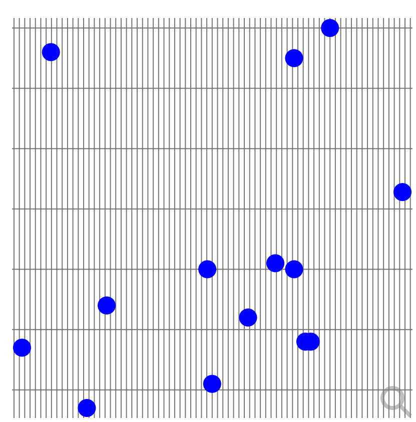
Prenatal Care Percentage	Health Expenditure Percentage of GDP
47.9	9.6
58.3	5.4
84.7	5.2
89.8	6.1
93.3	6
93.7	5.2
100	10
93.3	9.5
78	4.1
96.4	4.8
95.4	4.8
54.6	3.7
42.5	4.7
77.1	6

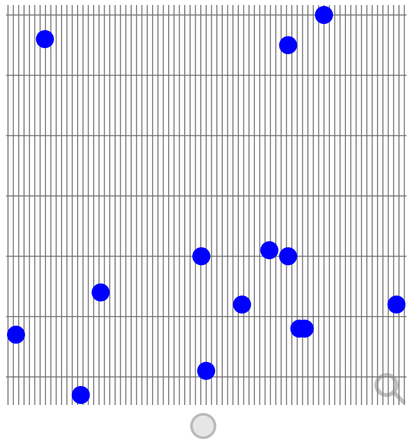
a) Since data were collected for 2 quantitative variable(s), the correct graph to make is a Scatter plot.

b) Make a scatterplot of X versus Y in StatCrunch or on your TI84. Which of the following is the correct graph?









c) Which of the following is true about the relation between Prenatal Care Percentage and Health Expenditure Percentage of GDP based on the graph above?

There doesn't appear to be a relation between the 2 Health

Select an answer

Question 22

0/1 pt 3 19

Katie recorded her bowling scores over the last 19 games. By looking at her data, she was able to get a better picture of her performances. Make a stem-leaf plot of her bowling scores and answer the questions:

~~151~~	~~147~~	~~158~~	~~148~~	~~141~~	~~154~~
~~129~~	~~108~~	~~120~~	~~136~~	~~140~~	~~141~~

Complete the stem-and-leaf plot below.

*** You must put commas between the **SORTED leaves least to greatest**.*
*** Type **"DNE"** if the leaves don't exist.*

Stems	Leaves
10	8
11	DNE
12	0 9
13	6
14	0 1 1 7 8
15	1 4 8

Key: 12|3 = 12|3

Answer above → (Shift 

*) for the separator bar with **NO Spaces**.*

Question 23 0/1 pt 3 19

If you randomly select a card from a well-shuffled standard deck of 52 cards, what is the probability that the card you select is a 9 or Jack? (Your answer must be in the form of a reduced fraction.)

4 cards of 9's
4 cards of Jack
8 cards → $\frac{8}{52}$

2/13

Question 24

0/1 pt 3 19

In a study of brand recognition, 1155 consumers knew of Google, and 7 did not. Use these results to estimate the probability that a randomly selected consumer will recognize Google.

1155+7 = 1162

Report your answer as a decimal accurate to 3 decimal places.

prob = 0.994

$\frac{1165}{1162} = 0.9939$

Question 25

0/1 pt 3 19

P(A and B) = P(A) × P(B)

Let `A` and `B` be independent events, such that `P(A)=0.5848` and `P(B)=0.5125` . Find the following probabilities:

0.5848 × 0.5125 = 0.299971

P(A and B)= 0.2997 (Round the answer to 4 decimals)

Question 26

0/1 pt 3 19

The number of credit cards in the wallet of a randomly selected person varies from 0 to 6. The probability distribution of the number of credit cards, `X`, is as follows:

`x_i`	0	1	2	3	4	5	6	Total
`P(X=x_i)`	0.01	0.13	0.16	0.43	0.15	0.11	0.01	1

1. Use the random variable notation to express symbolically each of the following:

   - X=3 An event in which the number of credit cards is exactly 3.

   -P(X=0) The probability that the number of credit cards is exactly 0.

   P(X=4) = 0.15 The probability that the number of credit cards is exactly 4 is equal to 0.15.

2. Use the above probability distribution table to find the following:
   a. `P(2<X<4)=` P(X=3) 0.43 (Round the answer to 2 decimals)
   b. `P(0<X<3)=` P(X=1) + P(X=2) → 0.13 + 0.16 0.29 (Round the answer to 2 decimals)
   c. `P(X>0)=` 1 − P(X=0) → 1 − 0.01 0.99 (Round the answer to 2 decimals)

Question 27

0/1 pt 3 19

A poll is given, showing 70% are in favor of a new building project.

If 9 people are chosen at random, what is the probability that exactly 2 of them favor the new building project?

0.0039.

Binomial → n = 9 p = 0.70 k = 2

$P(X=2) = \binom{9}{2} \times (0.70)^2 \times (0.30)^{9-7}$

↓

$36 \times 0.49 \times 0.0002187 = 0.0039$